

B.E. ELECTRICAL ENGINEERING
SECOND YEAR
SECOND SEMESTER EXAM 2024
FIELD THEORY
PART-II

Solved Answer Script in Academic Article Style

Time: 3 hours

Use separate answer-script for each part

Full Marks: 100

50 marks for each part

Answer any three questions. Two marks are for neatness. All symbols have their usual significance.

1. (a) What is the definition of curl? Derive the expression for curl of a vector field $\mathbf{C}$ using definition of curl, written as $\nabla \times \mathbf{C}$ where ∇ is the vector differential operator in Cartesian co-ordinates and explain its physical significance.

Short answer-building plan

First define curl as circulation per unit area. Then take a small rectangular loop in the xy -plane. Find line integral of $\mathbf{C}$ around four sides. Divide by area $dx dy$ to get the z -component of curl. Then write the full Cartesian expression and finally explain physical significance.

Detailed semester-style answer

Curl of a vector field gives the local rotational tendency or circulation tendency of the field at a point.

For a vector field $\mathbf{C}$, the component of curl along unit normal $\hat{\mathbf{n}}$ is defined as

$$(\nabla \times \mathbf{C}) \cdot \hat{\mathbf{n}} = \lim_{\Delta S \rightarrow 0} \frac{1}{\Delta S} \oint_C \mathbf{C} \cdot d\mathbf{l}$$

where C is the closed boundary of the small surface ΔS .

Let

$$\mathbf{C} = C_x \hat{\mathbf{x}} + C_y \hat{\mathbf{y}} + C_z \hat{\mathbf{z}}.$$

To derive the z -component of curl, consider a small rectangular path in the xy -plane.

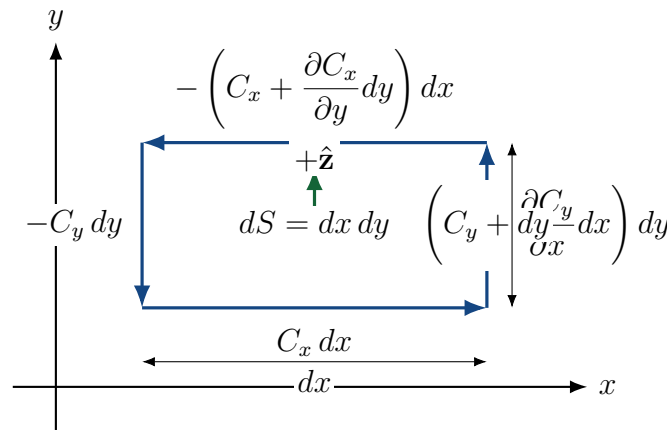


Figure 1: Small rectangular path in the xy -plane for deriving the z -component of curl.

Detailed semester-style answer

Now,

$$\oint \mathbf{C} \cdot d\mathbf{l} = \text{sum of line integrals over the four sides.}$$

For side 1:

$$\int_1 \mathbf{C} \cdot d\mathbf{l} = C_x dx.$$

For side 2:

$$\int_2 \mathbf{C} \cdot d\mathbf{l} = \left(C_y + \frac{\partial C_y}{\partial x} dx \right) dy.$$

For side 3:

$$\int_3 \mathbf{C} \cdot d\mathbf{l} = - \left(C_x + \frac{\partial C_x}{\partial y} dy \right) dx.$$

For side 4:

$$\int_4 \mathbf{C} \cdot d\mathbf{l} = -C_y dy.$$

Adding,

$$\oint \mathbf{C} \cdot d\mathbf{l} = C_x dx + \left(C_y + \frac{\partial C_y}{\partial x} dx \right) dy - \left(C_x + \frac{\partial C_x}{\partial y} dy \right) dx - C_y dy.$$

After cancellation,

$$\oint \mathbf{C} \cdot d\mathbf{l} = \left(\frac{\partial C_y}{\partial x} - \frac{\partial C_x}{\partial y} \right) dx dy.$$

Therefore,

$$(\nabla \times \mathbf{C})_z = \frac{\partial C_y}{\partial x} - \frac{\partial C_x}{\partial y}$$

Similarly,

$$(\nabla \times \mathbf{C})_x = \frac{\partial C_z}{\partial y} - \frac{\partial C_y}{\partial z}$$

and

$$(\nabla \times \mathbf{C})_y = \frac{\partial C_x}{\partial z} - \frac{\partial C_z}{\partial x}$$

Final result / exam point

Thus,

$$\nabla \times \mathbf{C} = \left(\frac{\partial C_z}{\partial y} - \frac{\partial C_y}{\partial z} \right) \hat{\mathbf{x}} + \left(\frac{\partial C_x}{\partial z} - \frac{\partial C_z}{\partial x} \right) \hat{\mathbf{y}} + \left(\frac{\partial C_y}{\partial x} - \frac{\partial C_x}{\partial y} \right) \hat{\mathbf{z}}$$

In determinant form,

$$\nabla \times \mathbf{C} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ C_x & C_y & C_z \end{vmatrix}$$

Physical significance: Curl measures the local circulation of a vector field. If $\nabla \times \mathbf{C} = 0$, the field is irrotational. If $\nabla \times \mathbf{C} \neq 0$, the field has rotational tendency. Direction of curl is given by the right-hand rule.

1. (b) Derive the $\nabla \cdot \mathbf{B}$ from definition of Divergence in any co-ordinates and show that $\nabla \cdot \mathbf{B} = 0$ where $\mathbf{B}$ is magnetic flux density.

Short answer-building plan

First write the fundamental definition of divergence as flux per unit volume. Then write the divergence expression for a general vector. Finally apply it to magnetic flux density $\mathbf{B}$ and explain why its divergence is zero.

Detailed semester-style answer

Divergence of a vector field is defined as the net outward flux per unit volume from a very small closed surface.

For vector field $\mathbf{B}$,

$$\nabla \cdot \mathbf{B} = \lim_{\Delta V \rightarrow 0} \frac{1}{\Delta V} \oint_S \mathbf{B} \cdot d\mathbf{S}$$

In Cartesian coordinates,

$$\mathbf{B} = B_x \hat{\mathbf{x}} + B_y \hat{\mathbf{y}} + B_z \hat{\mathbf{z}}.$$

Using a small rectangular volume $dx dy dz$, the net outward flux through opposite faces gives:

$$\oint_S \mathbf{B} \cdot d\mathbf{S} = \left(\frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} + \frac{\partial B_z}{\partial z} \right) dx dy dz.$$

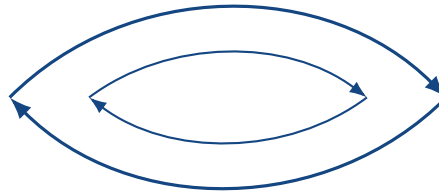
Dividing by volume,

$$\Delta V = dx dy dz,$$

we get

$$\nabla \cdot \mathbf{B} = \frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} + \frac{\partial B_z}{\partial z}$$

$\mathbf{B}$ lines are closed



No magnetic source or sink

Figure 2: Magnetic flux lines form closed loops.

Detailed semester-style answer

For magnetic flux density, Gauss's law for magnetism is

$$\oint_S \mathbf{B} \cdot d\mathbf{S} = 0$$

Using divergence theorem,

$$\oint_S \mathbf{B} \cdot d\mathbf{S} = \iiint_V \nabla \cdot \mathbf{B} dV.$$

Therefore,

$$\iiint_V \nabla \cdot \mathbf{B} dV = 0.$$

Since the volume V is arbitrary,

$$\nabla \cdot \mathbf{B} = 0$$

Final result / exam point

$$\nabla \cdot \mathbf{B} = 0$$

means:

- magnetic flux lines never start or end,
- magnetic lines form closed loops,
- isolated magnetic monopoles do not exist,
- net magnetic flux through any closed surface is zero.

2. (a) Establish the Stoke's Theorem.

Short answer-building plan

State Stoke's theorem first. Then explain the idea by dividing an open surface into many small elements. Internal boundary contributions cancel. Only the outer boundary remains, giving the theorem.

Detailed semester-style answer

Stoke's theorem states that the line integral of a vector field around a closed boundary is equal to the surface integral of the curl of that vector field over the open surface bounded by that boundary.

Mathematically,

$$\oint_C \mathbf{A} \cdot d\mathbf{l} = \iint_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S}$$

where C is the closed boundary and S is the open surface bounded by C .

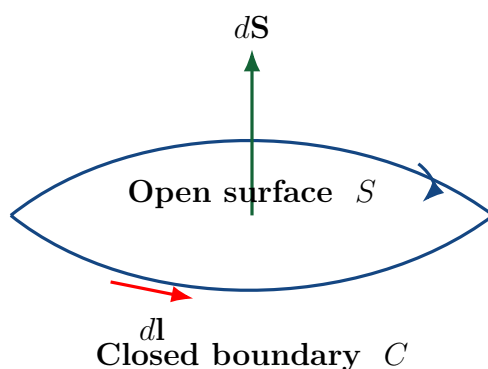


Figure 3: Stoke's theorem connects boundary circulation with surface curl.

Detailed semester-style answer

For a small surface element,

$$(\nabla \times \mathbf{A}) \cdot d\mathbf{S}$$

represents the circulation of $\mathbf{A}$ around the boundary of that small element.

Now divide the total open surface S into a large number of small surface elements.

When the line integrals around all small elements are added, the line integrals along internal common boundaries cancel because they are taken in opposite directions.

Only the line integral along the outer boundary C remains.

Therefore,

$$\sum (\nabla \times \mathbf{A}) \cdot d\mathbf{S} = \oint_C \mathbf{A} \cdot d\mathbf{l}.$$

In the limiting case,

$$\iint_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S} = \oint_C \mathbf{A} \cdot d\mathbf{l}$$

Thus Stoke's theorem is established.

2. (b) Establish $\nabla \times \mathbf{H} = \mathbf{J}$.

Short answer-building plan

Start with Ampere's circuital law in integral form. Write enclosed current in terms of current density. Apply Stoke's theorem to convert line integral of $\mathbf{H}$ into surface integral of $\nabla \times \mathbf{H}$. Then compare integrands.

Detailed semester-style answer

Ampere's circuital law states that the line integral of magnetic field intensity $\mathbf{H}$ around a closed path is equal to the total current enclosed by that path.

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = I_{\text{enc}}$$

The enclosed current may be written as

$$I_{\text{enc}} = \iint_S \mathbf{J} \cdot d\mathbf{S}$$

Therefore,

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = \iint_S \mathbf{J} \cdot d\mathbf{S}.$$

Using Stoke's theorem,

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = \iint_S (\nabla \times \mathbf{H}) \cdot d\mathbf{S}.$$

Thus,

$$\iint_S (\nabla \times \mathbf{H}) \cdot d\mathbf{S} = \iint_S \mathbf{J} \cdot d\mathbf{S}.$$

Since the surface is arbitrary,

$$\nabla \times \mathbf{H} = \mathbf{J}$$

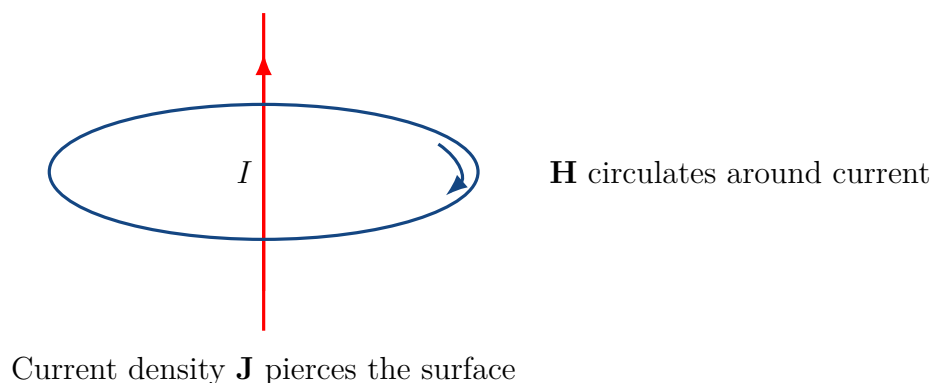


Figure 4: Physical view of $\nabla \times \mathbf{H} = \mathbf{J}$.

Final result / exam point

$$\nabla \times \mathbf{H} = \mathbf{J}$$

means current density is the source of circulation of magnetic field intensity.

2. (c) Establish $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$ and show the pictorial view of this relation.

Short answer-building plan

Start from Faraday's law in integral form. Use magnetic flux through an open surface. Apply Stoke's theorem on the electric field line integral. Compare the surface integrals to get the differential form.

Detailed semester-style answer

Faraday's law states that a time-varying magnetic flux produces an induced electric field. In integral form:

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = -\frac{d}{dt} \iint_S \mathbf{B} \cdot d\mathbf{S}$$

If the surface is stationary,

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = -\iint_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S}$$

By Stoke's theorem,

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = \iint_S (\nabla \times \mathbf{E}) \cdot d\mathbf{S}.$$

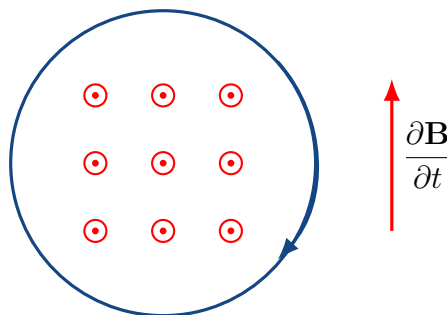
Hence,

$$\iint_S (\nabla \times \mathbf{E}) \cdot d\mathbf{S} = -\iint_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S}.$$

Since the surface is arbitrary,

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

Changing magnetic field



Induced circulating electric field $\mathbf{E}$

Figure 5: Changing magnetic field creates circulating electric field.

Final result / exam point

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

The negative sign represents Lenz's law.

3. (a) Establish Boundary Conditions for magnetic field when it passes through two different magnetic media having permeability μ_1 and μ_2 respectively and show one example for application of this relation.

Short answer-building plan

Draw the interface between two magnetic media. Apply Ampere's law to a small rectangular loop to get tangential H condition. Apply Gauss's law for magnetism to a small pillbox to get normal B condition. Combine both to get magnetic refraction relation.

Detailed semester-style answer

Consider two magnetic media separated by a plane boundary.

Let the permeabilities be

$$\mu_1 \quad \text{and} \quad \mu_2.$$

Also,

$$\mathbf{B}_1 = \mu_1 \mathbf{H}_1, \quad \mathbf{B}_2 = \mu_2 \mathbf{H}_2$$

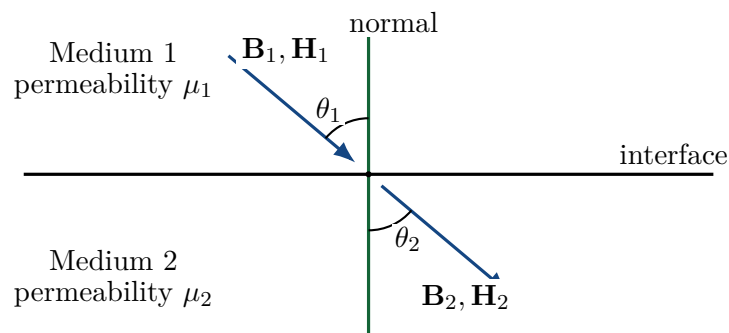


Figure 6: Magnetic field passing through two magnetic media without surface current at the interface.

Detailed semester-style answer

Condition 1: Tangential component of $\mathbf{H}$

Apply Ampere's circuital law to a very small rectangular loop across the boundary:

$$\oint \mathbf{H} \cdot d\mathbf{l} = I_{\text{enc}}.$$

If there is no surface current at the boundary,

$$I_{\text{enc}} = 0.$$

Then,

$$H_{1t} \Delta l - H_{2t} \Delta l = 0.$$

So,

$$H_{1t} = H_{2t}$$

If θ_1 and θ_2 are measured from the normal, then

$$H_1 \sin \theta_1 = H_2 \sin \theta_2$$

Condition 2: Normal component of B

Gauss's law for magnetism is

$$\oint_S \mathbf{B} \cdot d\mathbf{S} = 0.$$

Apply it to a very small pillbox across the boundary:

$$B_{1n}\Delta S - B_{2n}\Delta S = 0.$$

Therefore,

$$B_{1n} = B_{2n}$$

If angles are measured from the normal:

$$B_1 \cos \theta_1 = B_2 \cos \theta_2$$

Since

$$B_1 = \mu_1 H_1, \quad B_2 = \mu_2 H_2,$$

we get

$$\mu_1 H_1 \cos \theta_1 = \mu_2 H_2 \cos \theta_2.$$

Using

$$H_1 \sin \theta_1 = H_2 \sin \theta_2,$$

we obtain

$$\frac{\tan \theta_1}{\tan \theta_2} = \frac{\mu_1}{\mu_2}$$

Final result / exam point

Boundary conditions for magnetic field are:

$$H_{1t} = H_{2t}$$

and

$$B_{1n} = B_{2n}$$

Magnetic refraction relation:

$$\frac{\tan \theta_1}{\tan \theta_2} = \frac{\mu_1}{\mu_2}$$

Application: In transformer cores and magnetic circuits, flux tends to pass through high-permeability material. This is why iron cores guide magnetic flux efficiently.

3. (b) Derive electromagnetic wave equation of Magnetic Field ($\nabla^2 \mathbf{H} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{H}}{\partial t^2}$) in free space; then obtain an analytical solution of the wave equation of Magnetic Field ($\mathbf{H}$) considering it as a plane wave and draw the wave propagation.

Short answer-building plan

Use Maxwell's equations in free space. Take curl of Ampere-Maxwell law and use Faraday's law. Use the vector identity for curl of curl. Then assume a plane wave propagating along $+x$ direction and solve for $H_z(x, t)$.

Detailed semester-style answer

In free space,

$$\sigma = 0, \quad \rho = 0, \quad \mu = \mu_0, \quad \epsilon = \epsilon_0.$$

Maxwell's curl equations are:

$$\nabla \times \mathbf{H} = \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

and

$$\nabla \times \mathbf{E} = -\mu_0 \frac{\partial \mathbf{H}}{\partial t}$$

Take curl of the first equation:

$$\nabla \times (\nabla \times \mathbf{H}) = \epsilon_0 \frac{\partial}{\partial t} (\nabla \times \mathbf{E}).$$

Substitute Faraday's law:

$$\nabla \times \mathbf{E} = -\mu_0 \frac{\partial \mathbf{H}}{\partial t}.$$

Therefore,

$$\nabla \times (\nabla \times \mathbf{H}) = -\mu_0 \epsilon_0 \frac{\partial^2 \mathbf{H}}{\partial t^2}.$$

Using vector identity:

$$\nabla \times (\nabla \times \mathbf{H}) = \nabla(\nabla \cdot \mathbf{H}) - \nabla^2 \mathbf{H}.$$

In free space,

$$\nabla \cdot \mathbf{B} = 0$$

and since

$$\mathbf{B} = \mu_0 \mathbf{H},$$

we get

$$\nabla \cdot \mathbf{H} = 0.$$

Hence,

$$-\nabla^2 \mathbf{H} = -\mu_0 \epsilon_0 \frac{\partial^2 \mathbf{H}}{\partial t^2}.$$

Thus,

$$\nabla^2 \mathbf{H} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{H}}{\partial t^2}$$

Detailed semester-style answer

Plane wave solution:

Let the wave propagate in the $+x$ direction and let magnetic field be along z direction:

$$\mathbf{H} = H_z(x, t) \hat{\mathbf{z}}$$

Then,

$$\nabla^2 \mathbf{H} = \frac{\partial^2 H_z}{\partial x^2} \hat{\mathbf{z}}.$$

So,

$$\frac{\partial^2 H_z}{\partial x^2} = \mu_0 \epsilon_0 \frac{\partial^2 H_z}{\partial t^2}$$

A sinusoidal travelling wave solution is

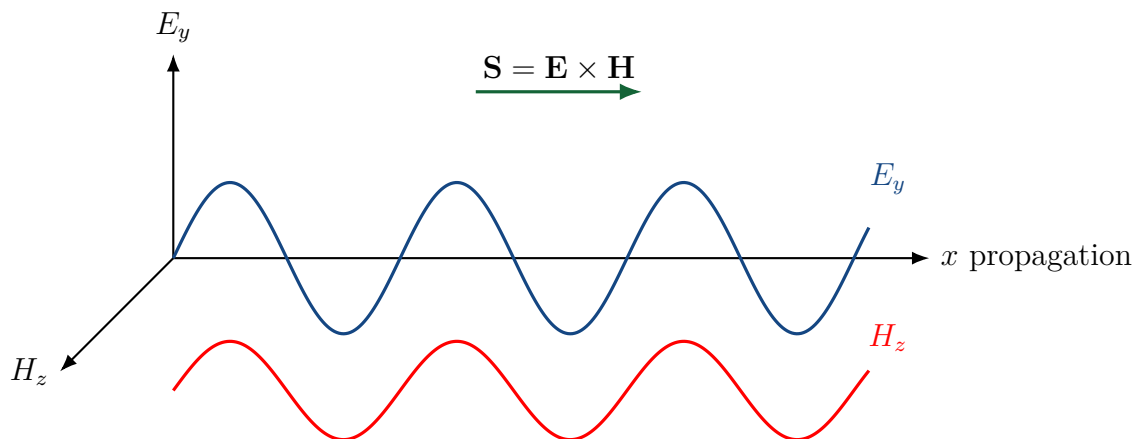
$$H_z(x, t) = H_0 \cos(\omega t - \beta x)$$

where

$$\beta = \omega \sqrt{\mu_0 \epsilon_0}$$

The wave velocity is

$$v = \frac{\omega}{\beta} = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$



Plane EM wave: $\mathbf{E}$, $\mathbf{H}$ and propagation direction are mutually perpendicular

Figure 7: Plane electromagnetic wave propagation in free space.

Final result / exam point

$$\nabla^2 \mathbf{H} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{H}}{\partial t^2}$$

For a plane wave:

$$H_z(x, t) = H_0 \cos(\omega t - \beta x)$$

where

$$\beta = \omega \sqrt{\mu_0 \epsilon_0}, \quad v = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

4. (a) Establish “Poynting Theorem”.

Short answer-building plan

First define Poynting vector. Then start from Maxwell’s curl equations. Take scalar product of Faraday’s law with $\mathbf{H}$ and Ampere-Maxwell law with $\mathbf{E}$. Use vector identity for divergence of cross product. Integrate over volume to obtain Poynting theorem.

Detailed semester-style answer

Poynting vector represents electromagnetic power flow per unit area.

It is defined as

$$\mathbf{S} = \mathbf{E} \times \mathbf{H}$$

Its unit is

$$\text{W/m}^2$$

Maxwell’s curl equations are:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

and

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}.$$

Take dot product of the first equation with $\mathbf{H}$:

$$\mathbf{H} \cdot (\nabla \times \mathbf{E}) = -\mathbf{H} \cdot \frac{\partial \mathbf{B}}{\partial t}.$$

Take dot product of the second equation with $\mathbf{E}$:

$$\mathbf{E} \cdot (\nabla \times \mathbf{H}) = \mathbf{E} \cdot \mathbf{J} + \mathbf{E} \cdot \frac{\partial \mathbf{D}}{\partial t}.$$

Using vector identity:

$$\nabla \cdot (\mathbf{E} \times \mathbf{H}) = \mathbf{H} \cdot (\nabla \times \mathbf{E}) - \mathbf{E} \cdot (\nabla \times \mathbf{H}).$$

Therefore,

$$\nabla \cdot (\mathbf{E} \times \mathbf{H}) = -\mathbf{H} \cdot \frac{\partial \mathbf{B}}{\partial t} - \mathbf{E} \cdot \mathbf{J} - \mathbf{E} \cdot \frac{\partial \mathbf{D}}{\partial t}.$$

For a linear medium,

$$\mathbf{B} = \mu \mathbf{H}, \quad \mathbf{D} = \epsilon \mathbf{E}.$$

Thus,

$$\mathbf{H} \cdot \frac{\partial \mathbf{B}}{\partial t} = \frac{\partial}{\partial t} \left(\frac{1}{2} \mu H^2 \right),$$

and

$$\mathbf{E} \cdot \frac{\partial \mathbf{D}}{\partial t} = \frac{\partial}{\partial t} \left(\frac{1}{2} \epsilon E^2 \right).$$

So,

$$\nabla \cdot (\mathbf{E} \times \mathbf{H}) = -\frac{\partial}{\partial t} \left[\frac{1}{2} \epsilon E^2 + \frac{1}{2} \mu H^2 \right] - \mathbf{E} \cdot \mathbf{J}.$$

Detailed semester-style answer

Let

$$u_{em} = \frac{1}{2}\epsilon E^2 + \frac{1}{2}\mu H^2$$

be the electromagnetic energy density.

Then,

$$\nabla \cdot \mathbf{S} = -\frac{\partial u_{em}}{\partial t} - \mathbf{E} \cdot \mathbf{J}.$$

Integrating over a volume V :

$$\int_V \nabla \cdot \mathbf{S} dV = -\frac{\partial}{\partial t} \int_V u_{em} dV - \int_V \mathbf{E} \cdot \mathbf{J} dV.$$

Using divergence theorem:

$$\oint_S \mathbf{S} \cdot d\mathbf{S} = -\frac{\partial}{\partial t} \int_V u_{em} dV - \int_V \mathbf{E} \cdot \mathbf{J} dV$$

Final result / exam point

Poynting theorem states that:

$$\text{Power leaving a volume} = \text{rate of decrease of stored EM energy} - \text{ohmic loss}$$

Mathematically,

$$\oint_S (\mathbf{E} \times \mathbf{H}) \cdot d\mathbf{S} = -\frac{\partial}{\partial t} \int_V \left(\frac{1}{2}\epsilon E^2 + \frac{1}{2}\mu H^2 \right) dV - \int_V \mathbf{E} \cdot \mathbf{J} dV$$

4. (b) A square loop measuring 1.5 m by 1.5 m carries a 7.5 A steady current, where the loop is in xz plane, using Biot-Savart law compute the B-field and Energy density at a point 0.35 m away on axis of the loop, the y -axis, in air material. Derive the formula used if any.

Short answer-building plan

Draw the square loop in the xz -plane and point on the y -axis. Use the axial magnetic field formula for a rectangular loop derived from Biot-Savart law. For a square loop, put half-side $a = 0.75\text{ m}$ and axial distance $y = 0.35\text{ m}$. Then calculate magnetic flux density B and energy density $w_m = B^2/(2\mu_0)$.

Detailed semester-style answer

Given:

$$\text{side of square loop} = 1.5\text{ m}$$

$$I = 7.5\text{ A}$$

$$y = 0.35\text{ m}$$

Half-side of square loop:

$$a = \frac{1.5}{2} = 0.75\text{ m}.$$

For a square loop in the xz -plane, the point is on the axis of the loop, i.e. on the y -axis. The magnetic field is along the axis of the loop.

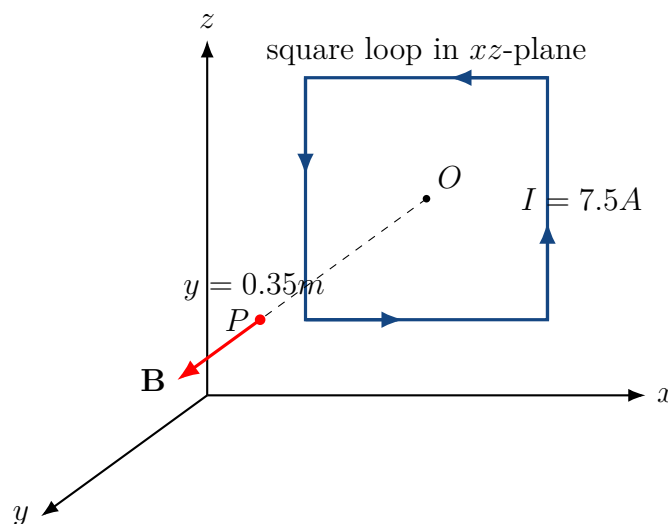


Figure 8: Square current loop in xz -plane and axial point on y -axis.

Detailed semester-style answer

For a rectangular loop with half sides a and b , the magnetic flux density at an axial point at

distance y is

$$B = \frac{\mu_0 I a b}{\pi \sqrt{a^2 + b^2 + y^2}} \left[\frac{1}{a^2 + y^2} + \frac{1}{b^2 + y^2} \right]$$

For a square loop,

$$a = b = 0.75 \text{ m.}$$

So,

$$B = \frac{2\mu_0 I a^2}{\pi (a^2 + y^2) \sqrt{2a^2 + y^2}}.$$

Now put:

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m}, \quad I = 7.5 \text{ A}, \quad a = 0.75 \text{ m}, \quad y = 0.35 \text{ m}.$$

$$B = \frac{2(4\pi \times 10^{-7})(7.5)(0.75)^2}{\pi [(0.75)^2 + (0.35)^2] \sqrt{2(0.75)^2 + (0.35)^2}}$$

$$(0.75)^2 = 0.5625, \quad (0.35)^2 = 0.1225.$$

$$a^2 + y^2 = 0.685.$$

$$2a^2 + y^2 = 2(0.5625) + 0.1225 = 1.2475.$$

$$\sqrt{2a^2 + y^2} = \sqrt{1.2475} \approx 1.1169.$$

Therefore,

$$B \approx 4.41 \times 10^{-6} \text{ T}.$$

Thus,

$$B \approx 4.41 \mu\text{T}$$

Energy density:

Magnetic energy density in air is

$$w_m = \frac{B^2}{2\mu_0}$$

$$w_m = \frac{(4.41 \times 10^{-6})^2}{2(4\pi \times 10^{-7})}$$

$$w_m \approx 7.74 \times 10^{-6} \text{ J/m}^3.$$

Final result / exam point

$$B \approx 4.41 \times 10^{-6} \text{ T}$$

$$w_m \approx 7.74 \times 10^{-6} \text{ J/m}^3$$

The magnetic field is directed along the axis of the square loop, i.e. along the y -axis.

5. (a) Establish $\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$ and show the pictorial view of this relation. What is displacement current?

Short answer-building plan

Start from old Ampere's law. Explain that it is incomplete for time-varying fields. Define displacement current density. Then write the complete Ampere-Maxwell equation and draw capacitor picture.

Detailed semester-style answer

For steady current,

$$\nabla \times \mathbf{H} = \mathbf{J}$$

But for time-varying electric fields, Maxwell added the displacement current density:

$$\mathbf{J}_d = \frac{\partial \mathbf{D}}{\partial t}$$

So total current density becomes

$$\mathbf{J}_{total} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

Therefore, complete Ampere-Maxwell law is

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

The integral form is

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = \iint_S \left(\mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \right) \cdot d\mathbf{S}$$

Using Stoke's theorem, the differential form is obtained as

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

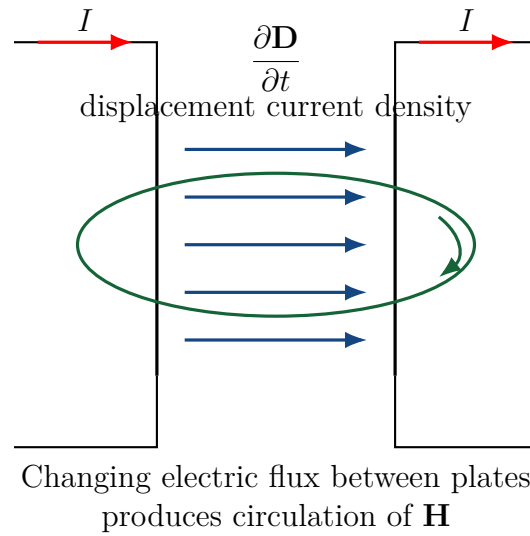


Figure 9: Pictorial view of Ampere-Maxwell law.

Final result / exam point

Displacement current density is

$$\mathbf{J}_d = \frac{\partial \mathbf{D}}{\partial t}$$

Displacement current through a surface is

$$I_d = \iint_S \frac{\partial \mathbf{D}}{\partial t} \cdot d\mathbf{S}$$

It makes Ampere's law valid for time-varying fields.

5. (b) What do you understand by electromagnetic wave polarization? Explain. Also calculate the speed of any electromagnetic wave in free space.

Short answer-building plan

First define polarization using direction of electric field vector. Then explain linear, circular and elliptical polarization. Finally calculate free-space speed using $c = 1/\sqrt{\mu_0\epsilon_0}$.

Detailed semester-style answer

Polarization of an electromagnetic wave describes the manner in which the electric field vector $\mathbf{E}$ varies with time at a fixed point in space.

Since an electromagnetic wave is transverse, the electric field is perpendicular to the direction of propagation.

The path traced by the tip of the electric field vector decides the type of polarization.

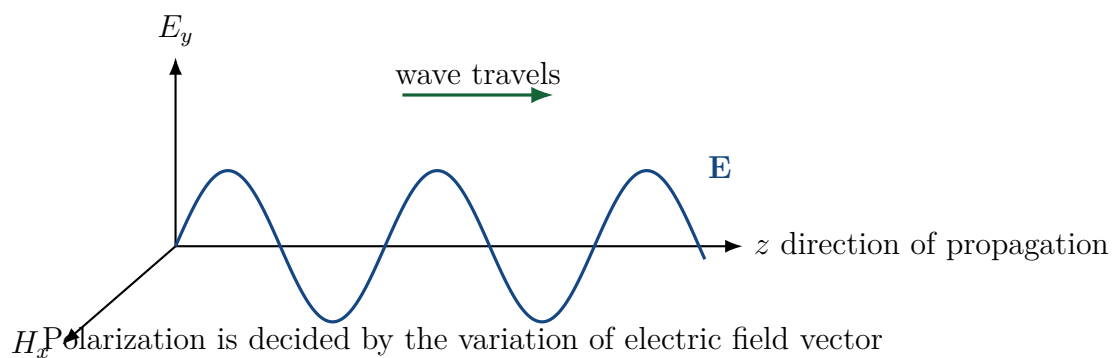


Figure 10: Electric field direction decides polarization of EM wave.

Detailed semester-style answer

1. Linear polarization:

If the electric field vector always remains along a fixed straight line, the wave is linearly polarized. Example:

$$\mathbf{E} = E_0 \cos(\omega t - \beta z) \hat{\mathbf{x}}$$

2. Circular polarization:

If two perpendicular electric field components have equal amplitude and phase difference of 90° , then the tip of the resultant electric field vector traces a circle.

Condition:

$$E_{0x} = E_{0y}, \quad \delta = \pm \frac{\pi}{2}$$

3. Elliptical polarization:

If the two perpendicular components have unequal amplitudes or any general phase difference, the tip of the resultant electric field vector traces an ellipse.

Elliptical polarization is the most general case. Linear and circular polarizations are special cases of elliptical polarization.

Detailed semester-style answer

Speed of EM wave in free space:

The velocity of an electromagnetic wave in a medium is

$$v = \frac{1}{\sqrt{\mu\epsilon}}$$

In free space,

$$\mu = \mu_0, \quad \epsilon = \epsilon_0.$$

So,

$$c = \frac{1}{\sqrt{\mu_0\epsilon_0}}$$

Now,

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$$

and

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}.$$

Therefore,

$$c = \frac{1}{\sqrt{(4\pi \times 10^{-7})(8.854 \times 10^{-12})}}$$

$$c \approx 3 \times 10^8 \text{ m/s}$$

Final result / exam point

Polarization describes the time variation of the electric field vector of an electromagnetic wave.

Types:

Linear, Circular and Elliptical Polarization

Free-space speed of EM wave:

$$c = \frac{1}{\sqrt{\mu_0\epsilon_0}} \approx 3 \times 10^8 \text{ m/s}$$